Newton School of Technology

1

Introduction to Computer Systems

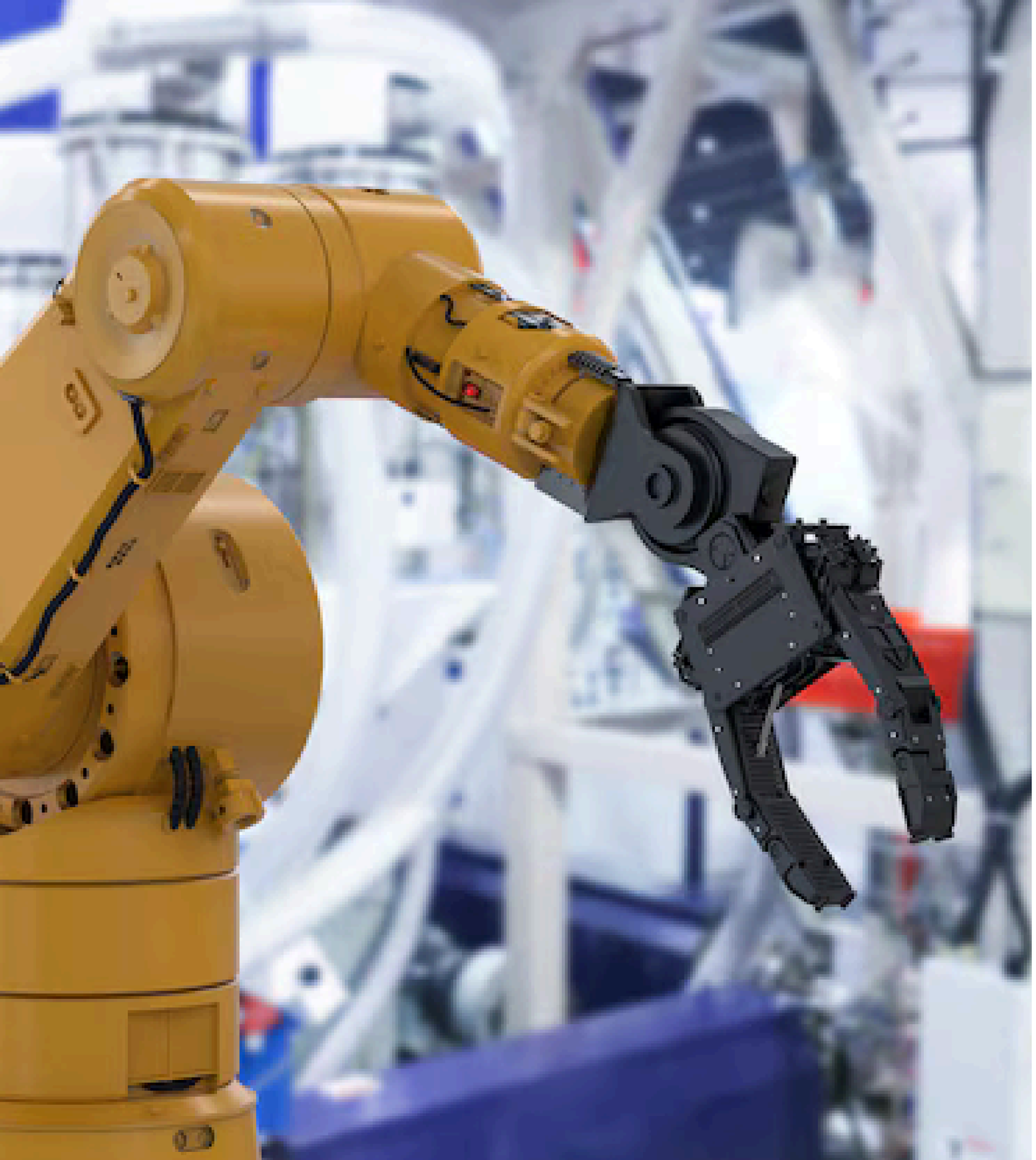
by Instructor Name

C01: System & AI Essentials

What is this?

What is this?

An MRI Machine



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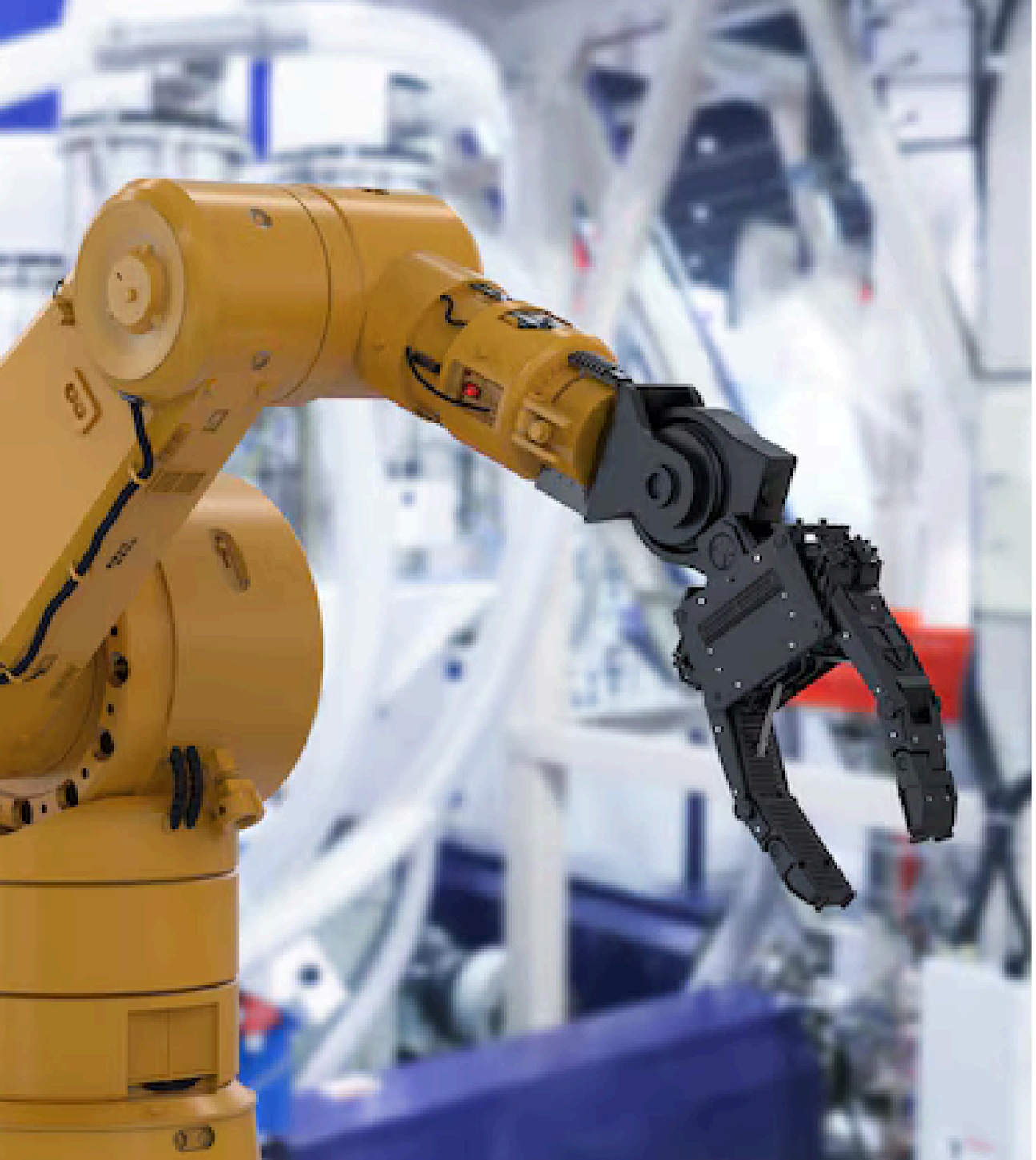
What is this?



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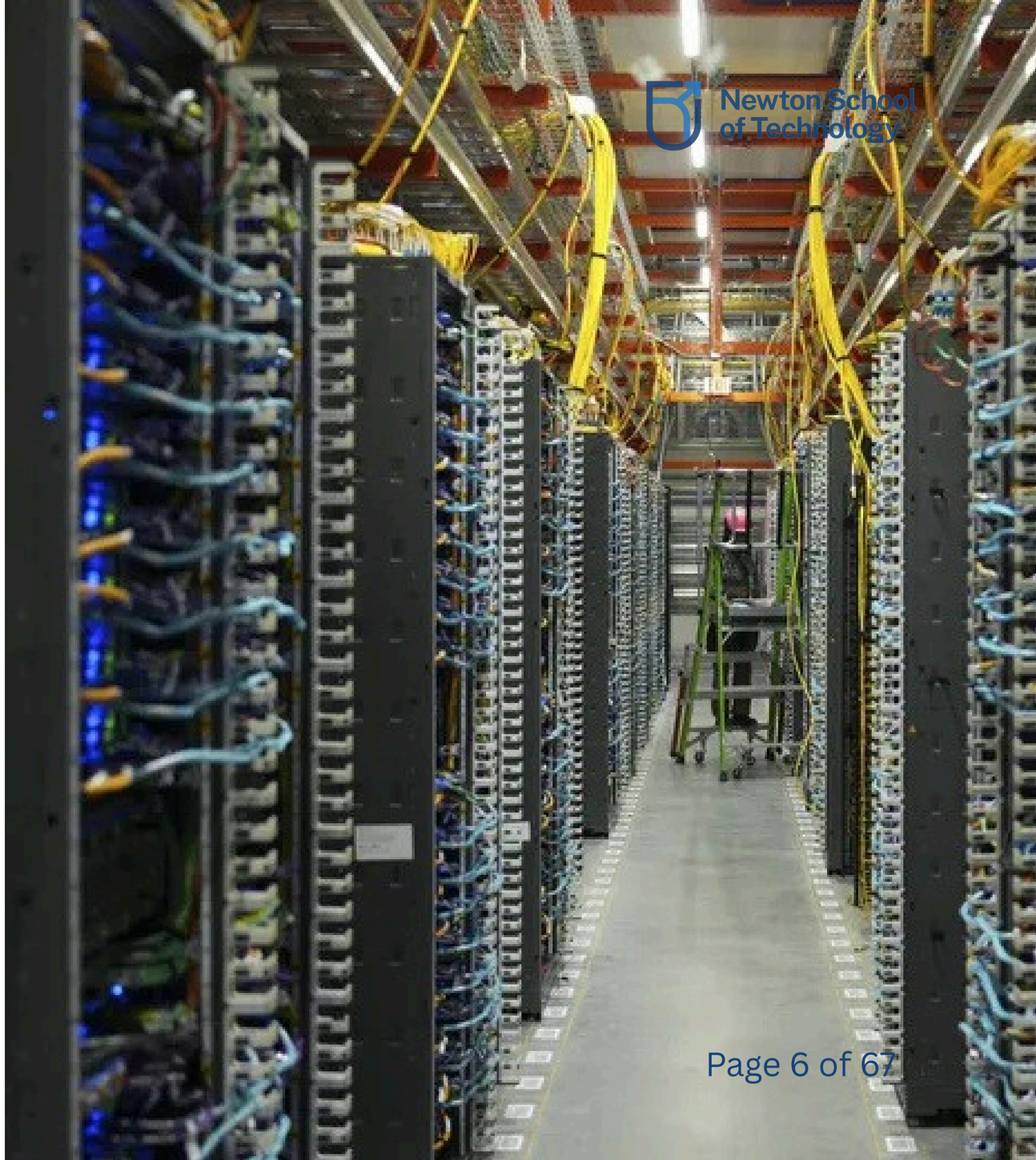
What is this?

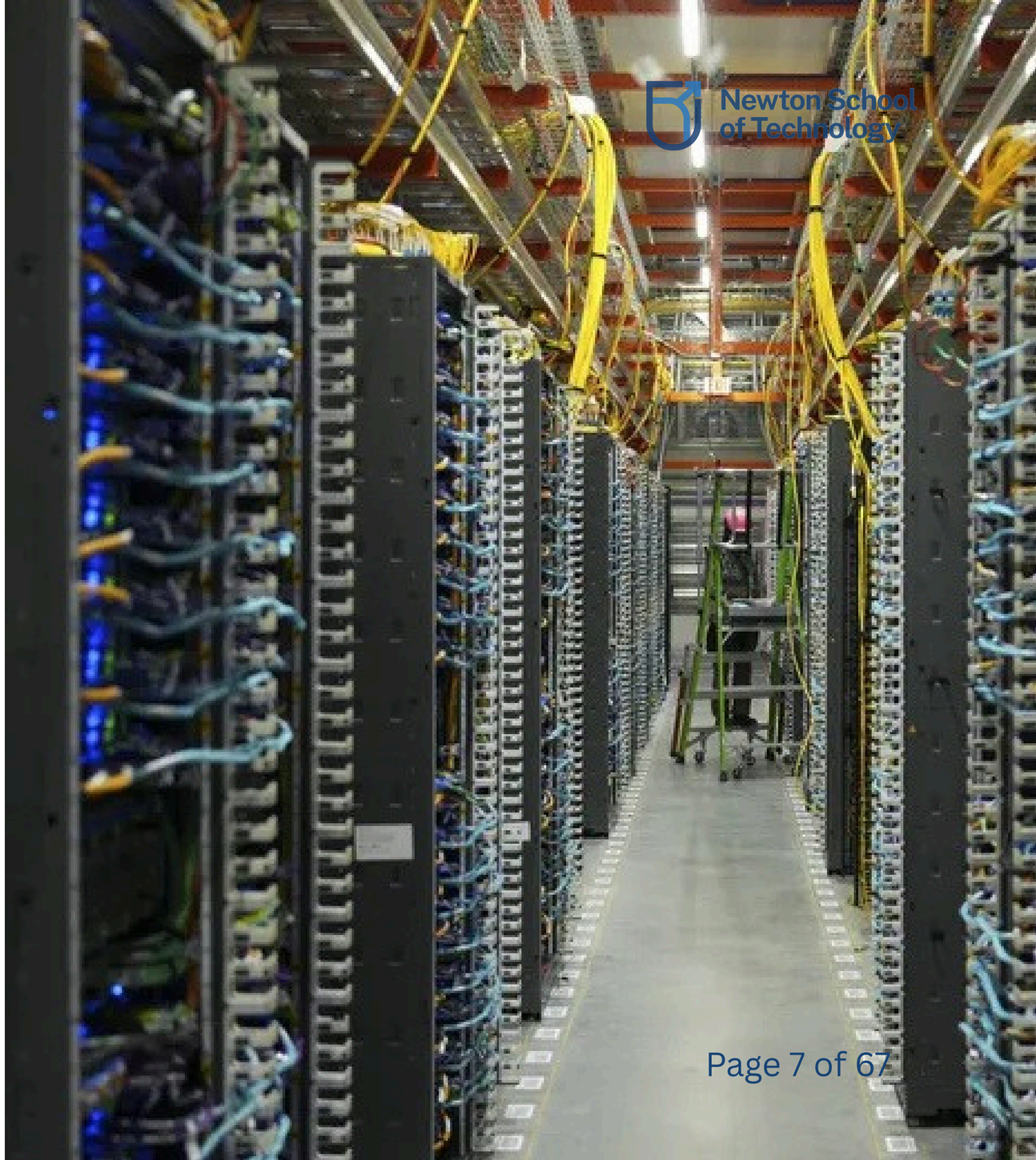
A Robotic Hand

What is this?



What is this?

A Data Center of servers



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What is this?

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What is this?

A Mobile Phone

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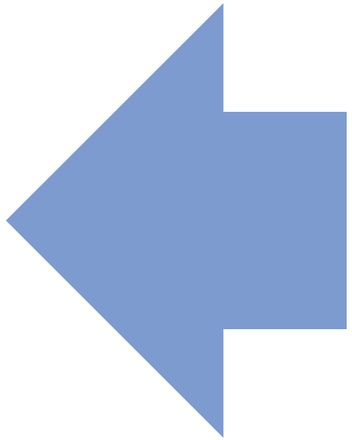
What is common in all these things?



All of these rely on computers

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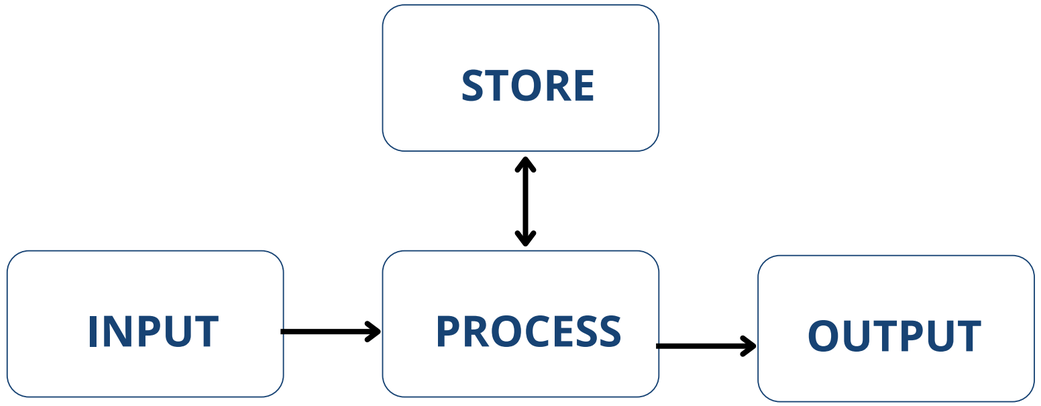
What is a Computer?

Computer

A computer is an electronic device that takes **input**, **processes** it, and gives **output** and sometimes **stores** information.

It follows a set of **instructions** to perform tasks like calculations, writing, drawing, playing music, and more.



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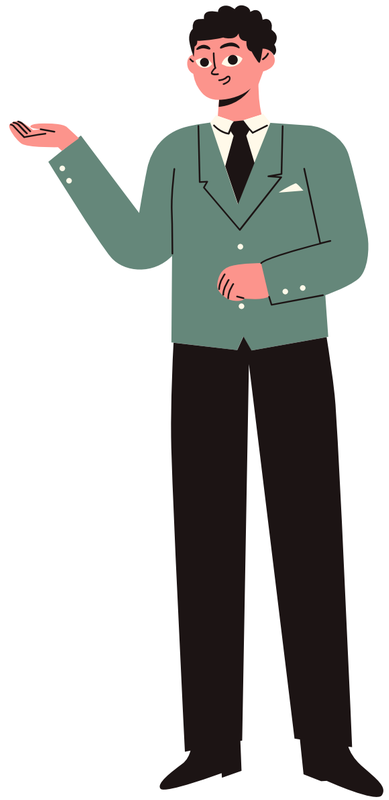
A juicer also takes input (oranges), processes it (crush) and produces and output (juice).

So does that mean a juicer is also a computer?

No!

Computers can do many tasks because they can run different programs. Non-computer devices do one specific job and can’t easily be changed to do something else.

The main difference is that **computers can be reprogrammed.**



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Can you imagine a day without computers?

A Day in the Life of Vishal

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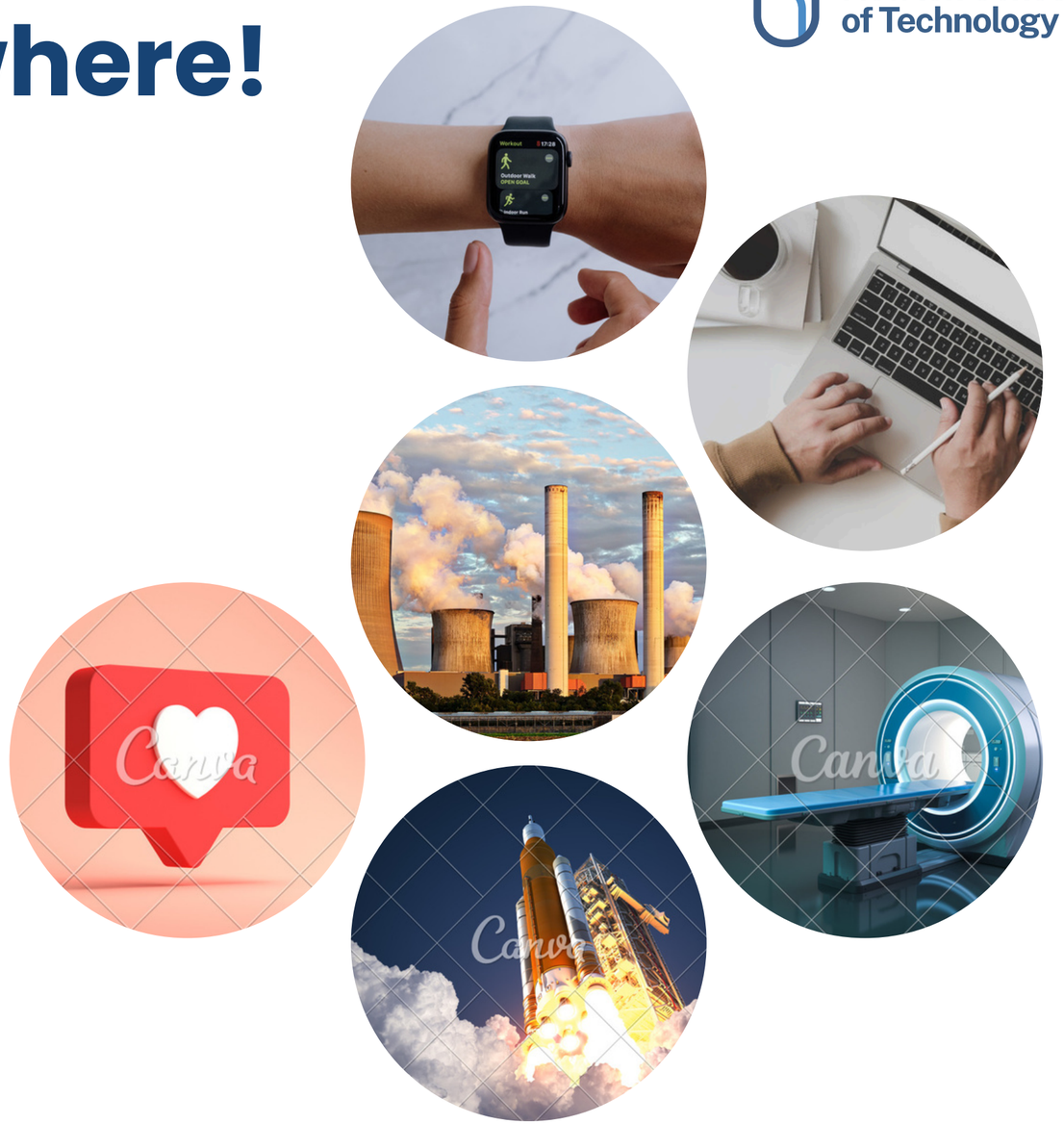
Computers are Everywhere!

Computers are everywhere in daily life

- Smartwatches monitor our health
- Smartphones keep us connected
- Laptops power work and learning

And they run the systems we depend on

- Critical infrastructure
- Space exploration
- Nuclear power plant operations
- Medical equipment, ensuring safety and precision



Computers in Healthcare

Modern healthcare depends on computers for monitoring, diagnosis, and life support.

- **Smartwatches** continuously monitor heart rate and health data.
- **MRI machines** use computers to reconstruct detailed body images.
- **Pacemakers** regulate heartbeats in real time.

Computers in Critical Infrastructure

Modern critical infrastructure depends on computers for safety, coordination, and real-time system management.

- **Nuclear power plants** rely on computerized safety systems.
- **Air traffic control** manages thousands of flights safely.
- **Power grids** balance electricity supply and demand every second.

Computers in Space Exploration

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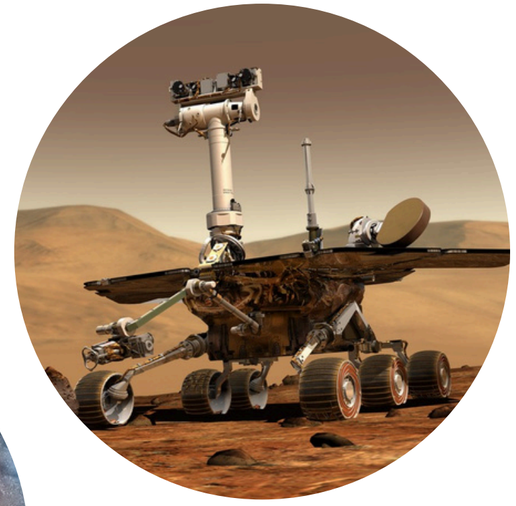
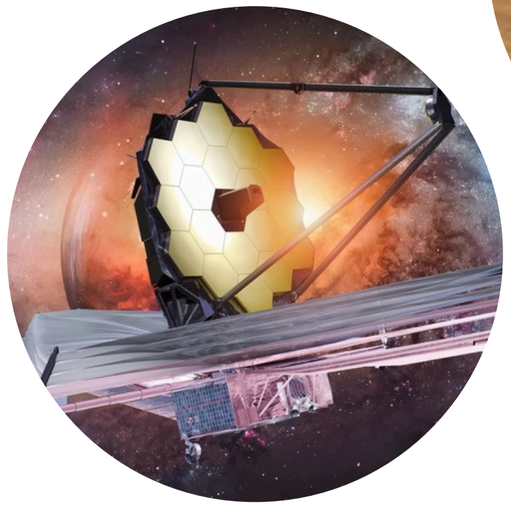
Modern space technology depends on computers for navigation, communication, data processing, and autonomous operations.

- **Mars rovers** operate using embedded computers.
- **James Webb Telescope** processes and transmits space data.
- **SpaceX rockets** land autonomously using software.





Computers at the Frontier of Innovation

Modern future technologies depend on computers for immersive experiences, automation, artificial intelligence, and innovation.

- **XR headsets** are bringing computers into everyday vision.
- **3D printers** turn digital designs into physical objects.
- **AI systems** can generate code, images, conversations, and scientific discoveries.

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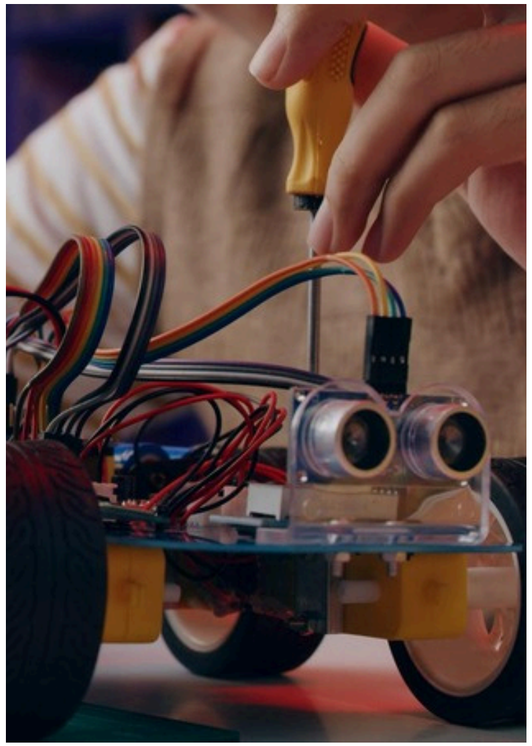
Inclass Quiz

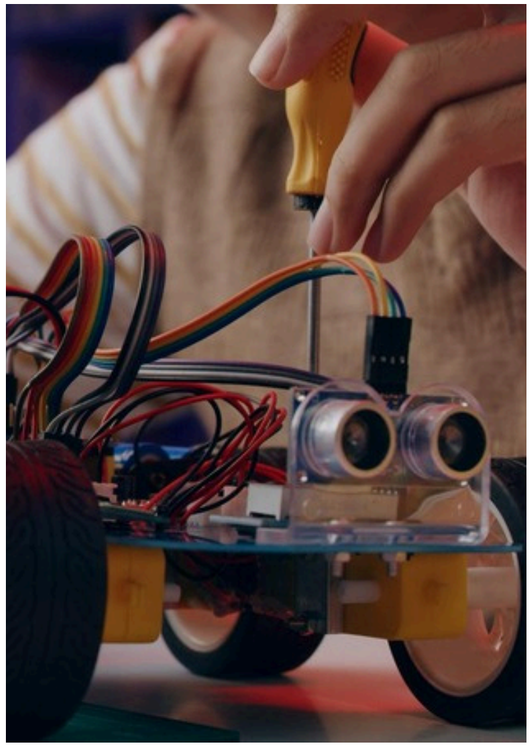
Types of Computers

Types of Computers



Embedded Systems

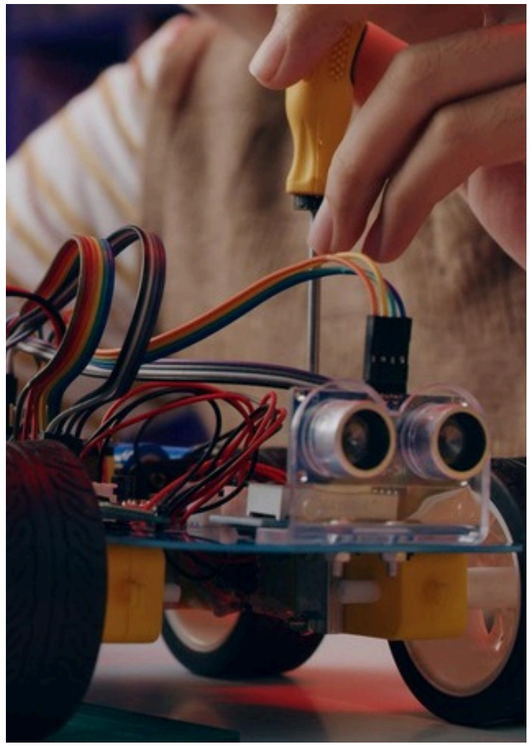
Types of Computers



Embedded Systems

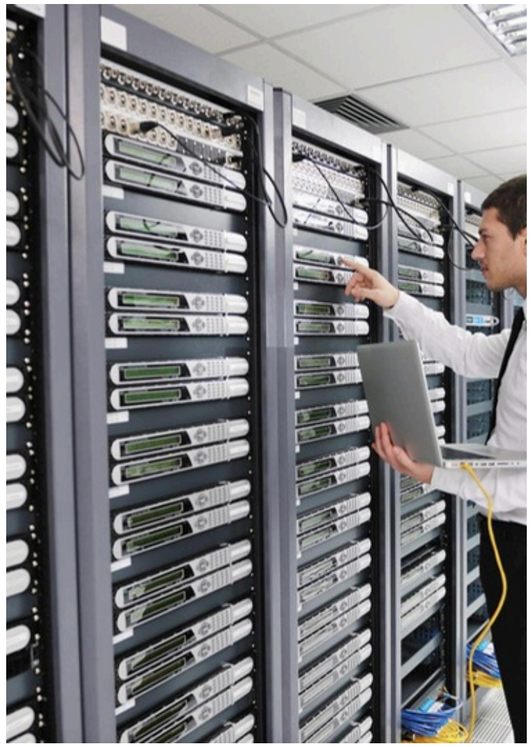
Personal Computer

Types of Computers



Embedded Systems

Personal Computer



Server & Datacenter

Types of Computers

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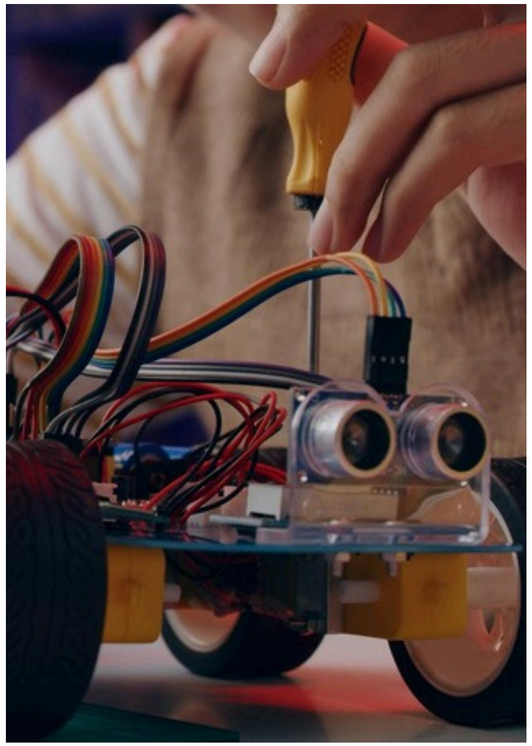
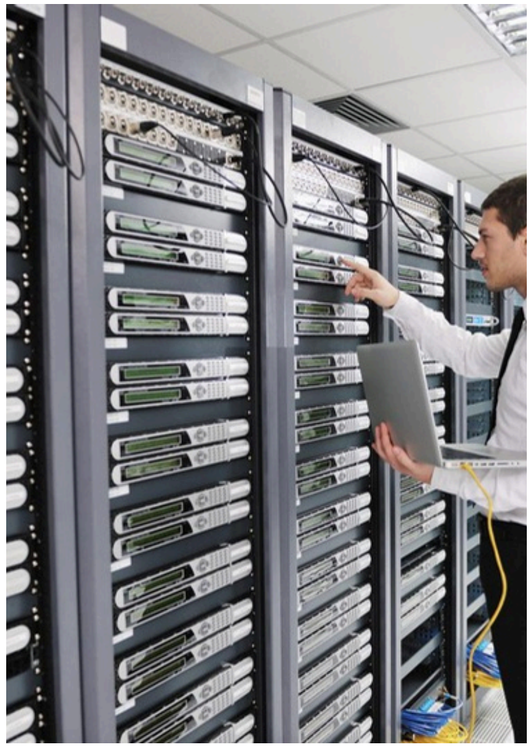


Embedded Systems

Personal Computer



Server & Datacenter

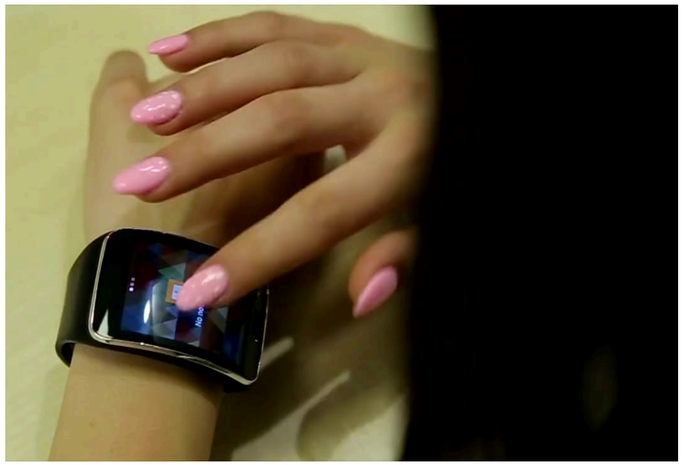
Supercomputer

Can you see a computer in these devices?

1

2

3



4

5

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Are these devices computers, or do they contain small computers that control their operation?

They contain small computers inside them. These are called **Embedded Systems**.

Let’s understand them!

Embedded Systems

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An embedded system is a **small computer built into a machine** or device to control how it works.

Assignment 1

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List out what are the functions of embedded computers in each of the following machines:

1. Washing Machine
2. Refrigerator
3. Smart Watch
4. Traffic Lights
5. Microwave Oven
6. ATM

Personal Computers

Personal computers (PCs) are computers designed for use by a single person, usually at home, school, or in the office.

- They include different types like **desktops**, **laptops**, and **smartphones**.
- People use personal computers for many everyday activities such as browsing the internet, writing documents, watching videos, playing games, and running various software programs.

Servers

Servers are computers that **store and provide data, websites, or services** to other computers through the internet or a network.

They help many users access information and applications at the same time.

Supercomputers

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Supercomputers are extremely **fast and powerful** computers used to **solve very large and complex problems**.

They are used for weather forecasting, scientific research, AI training, space research.

IBM's **Blue Gene/P** Supercomputer

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Inclass Quiz

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Innovations in Computing

Computers have evolved far beyond simple machines used for calculations and typing. In the last two decades, advancements in computing have changed the way we live, work, communicate, and learn.

Today, computers power technologies like **Artificial Intelligence (AI), Virtual Reality (VR), robots, social media, and 3D printing**.

Let’s explore some exciting technological advancements made possible by computers.

Artificial Intelligence

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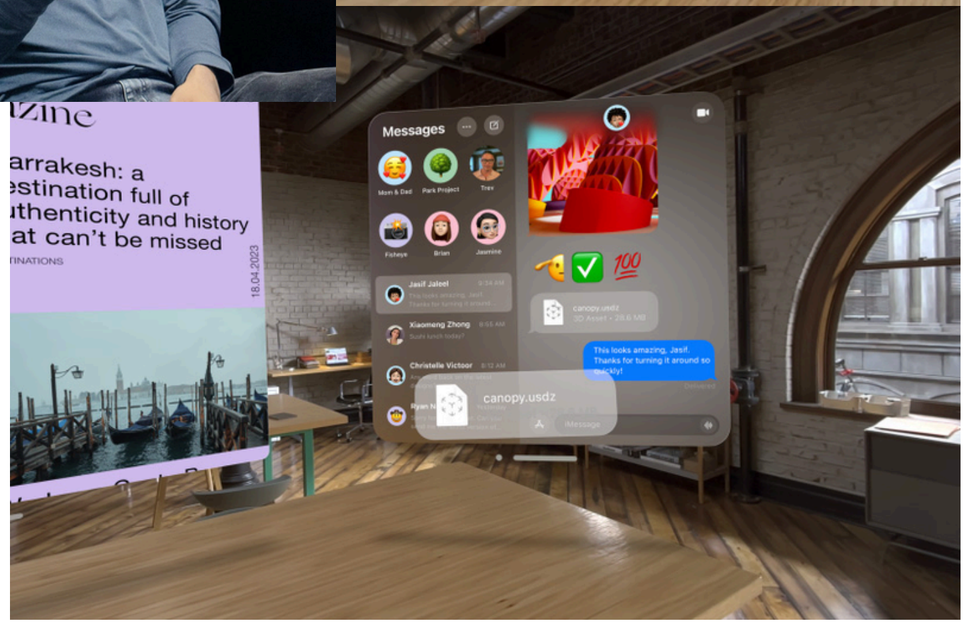
Artificial Intelligence (AI) enables computers to learn, make decisions, and perform tasks that normally require human intelligence.

Today, AI can do conversations, create images, write code, and help solve complex real-world problems.

Virtual Reality

Augmented Reality (AR) and **Virtual Reality (VR)** are immersive technologies that transform how we interact with the digital world.
AR overlays digital content onto the real world, while VR creates entirely virtual environments for users to explore.



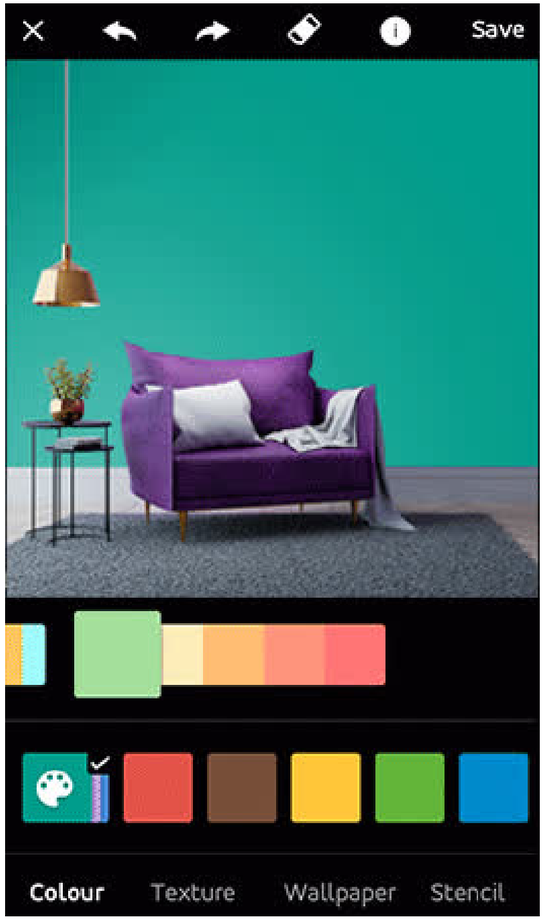
Apple Vision Pro is Apple's cutting-edge mixed reality headset, blending AR and VR to create an immersive digital experience.

Credits: MKBHD

Virtual Reality Examples

The **metaverse** is a collective virtual shared space that merges physical and digital realities, accessible through VR, AR, and other immersive technologies.

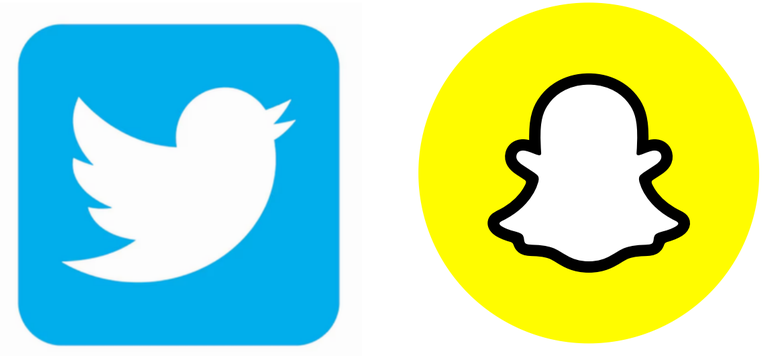
VR can be used to check out what certain paint colors would look like in your own home.



Social Media

Social media platforms like **Instagram, Facebook, and X** connect billions of people worldwide through instant communication and content sharing.

Modern social media uses smart algorithms for personalized feeds, recommendations, targeted advertisements, and real-time engagement, transforming communication, entertainment, and digital marketing.



Robots

A robot is a machine controlled by computers that can **sense, think, and perform actions automatically**.

Robots use **sensors** to observe the environment, **computers** to process information and make decisions, and motors to move and perform actions.

Today, robots are used in factories, hospitals, warehouses, space missions, and self-driving cars.

Quantum Computing

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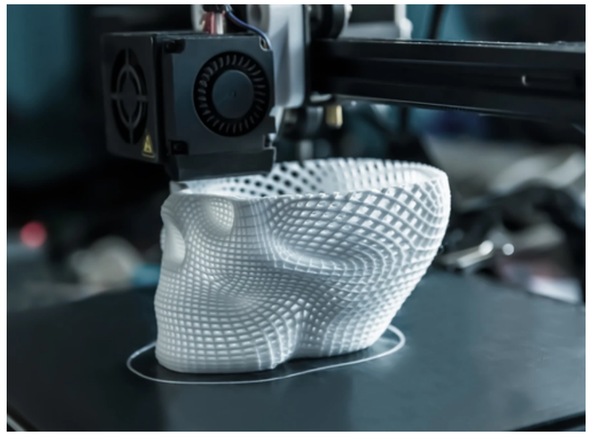
Quantum Computing is a type of computing that **uses the principles of quantum physics** to solve certain problems much faster than traditional computers.

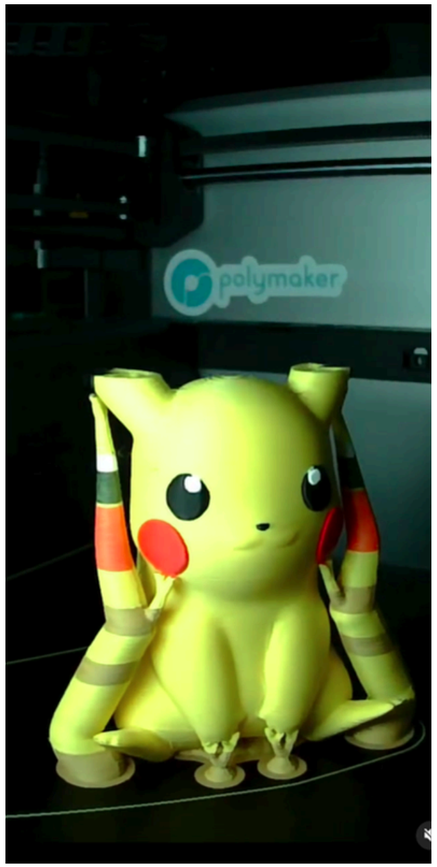
Traditional computers process one possibility at a time, while quantum computers can process many possibilities simultaneously.

3D Printing

3D printing is an innovative manufacturing technology that **creates physical objects** from digital designs by layering materials like plastic, metal, or resin.

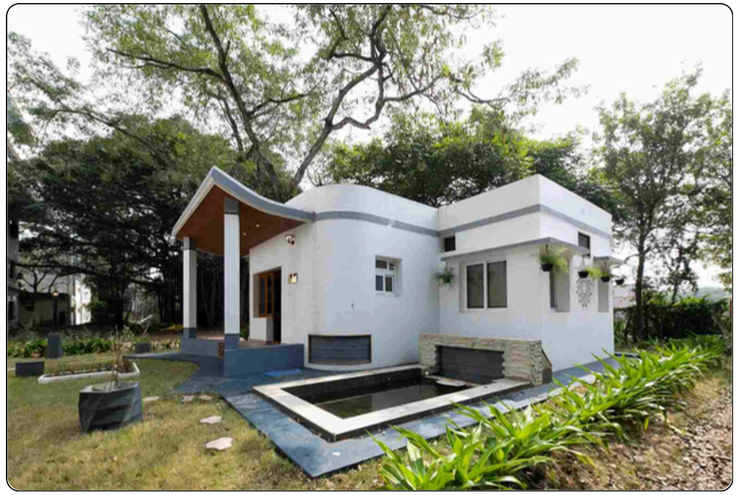
3D printing is used across various industries, including healthcare for printing prosthetics, architecture for creating detailed models, and even in space exploration for producing parts on-demand.





Credits: Polymaker

3D Printed Buildings



IIT Madras recently built India's first 3D-printed house and assembled it at the site. The entire project (500 sq. ft.) was completed in **21 days**.

Icon, a leader in large-scale 3D printing, recently announced a partnership with the home-building company Lennar to build the **largest community of 3D-printed homes** in the world in Austin, Texas.

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Inclass Quiz

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Appendix

Specialized Computers

Not all computers are designed for the same purpose. Different types of computers are built and optimized for specific kinds of work.

Types of Specialized Computers

- Embedded Computers
- Supercomputers
- Quantum Computers
- Servers

Quantum Computers

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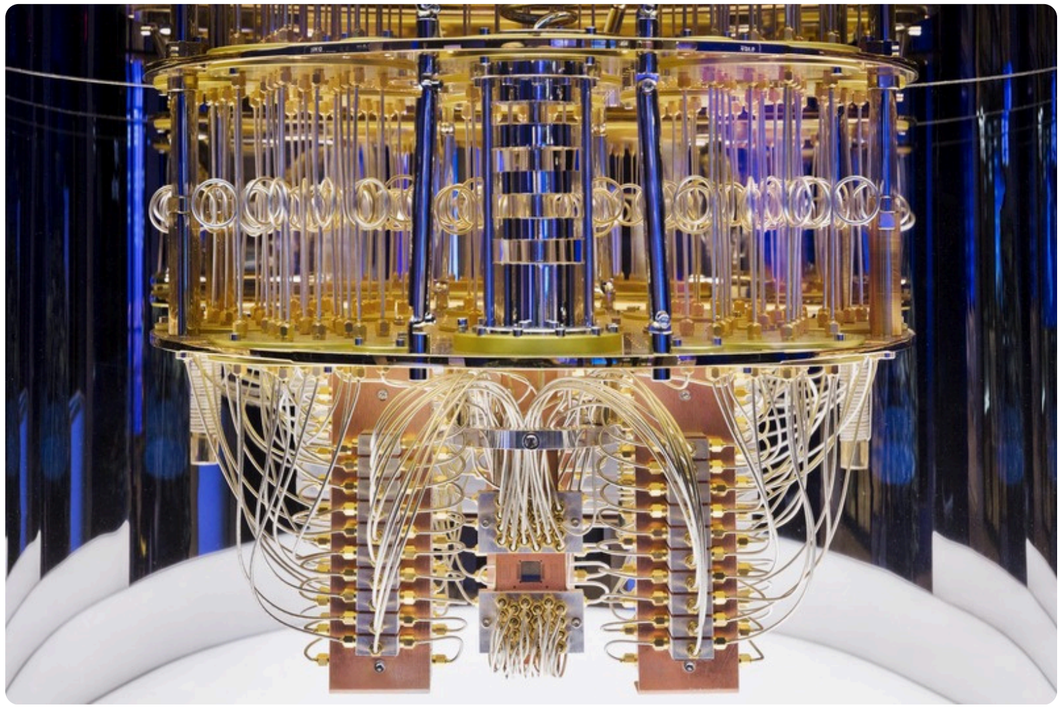
A quantum computer is an advanced computing system that leverages the principles of quantum physics to solve highly complex problems that are beyond the capabilities of even the most powerful classical supercomputers.



Reference : https://www.ibm.com/think/topics/quantum-computing

The CPU

The **Central Processing Unit** is the *brain* of the computer. It **executes instructions** from programs and **performs calculations**.

The **CPU** is installed on the **Motherboard** inside the CPU **Cabinet.**

Reference: https://www.ibm.com/think/topics/central-processing-unit

The GPU

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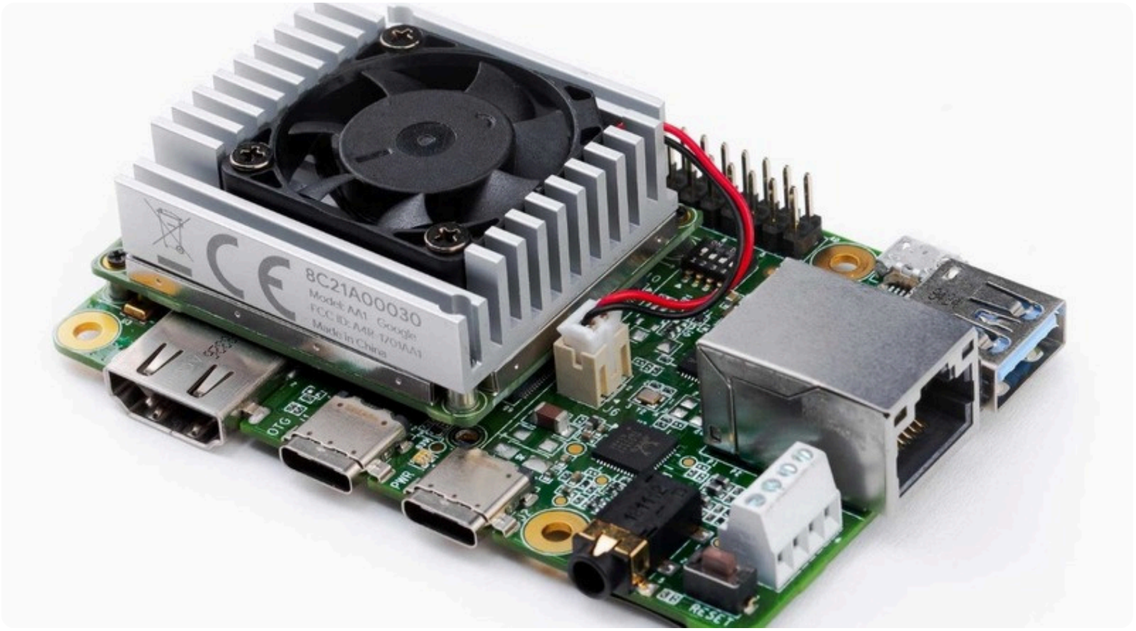
The **Graphics Processing Unit** (GPU) in your device helps handle **graphics-related work** like graphics, effects, and videos.

Reference : https://www.intel.com/content/www/us/en/products/docs/processors/what-is-a-gpu.html

The TPU

Tensor Processing Unit is a custom-designed processor developed by Google mainly for artificial intelligence (AI) and machine learning tasks.



Reference : https://cloud.google.com/tpu

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Quiz

Is Calculator a Computer?

Is Calculator a Computer?

Calculators are special-purpose computers designed for a narrow set of tasks (math operations), with fixed or limited programming, a simple interface, and minimal memory.

Yes, a calculator is a computer because it process input, perform calculations, and produce output.

What is Roomba?

What is Roomba?

A Roomba is a **robotic** vacuum cleaner. It is an embedded computer system because it contains a small computer designed for one specific job.

In a Roomba, the computer controls:

- sensors,
- wheels and motors,
- navigation,
- and cleaning operations.

Is a Tesla car one computer or many computers working together?

Is a Tesla car one computer or many computers working together?

A Tesla car contains **many small computers** working together.

Different computers inside the car control navigation, cameras, battery systems, braking, entertainment, and self-driving features.

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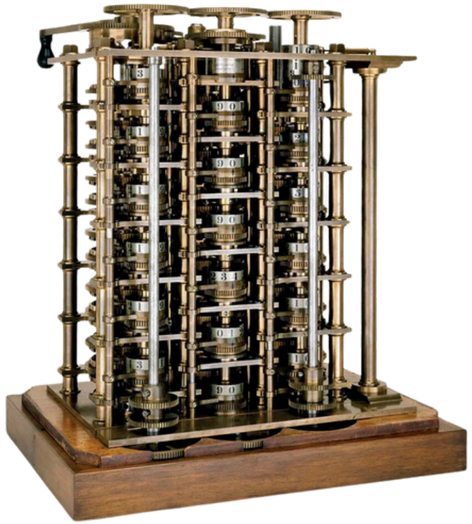
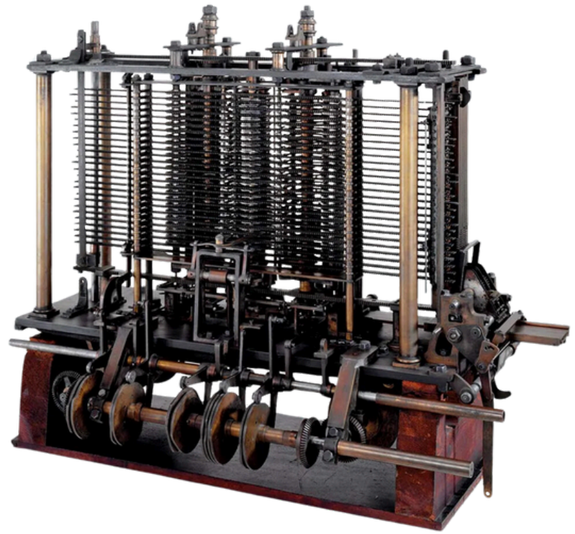
History of Computing

First Computer

Charles Babbage, a 19th-century mathematician and inventor, is often called the **"father of the computer"** for his design of the Analytical Engine a mechanical computer envisioned to perform complex calculations.

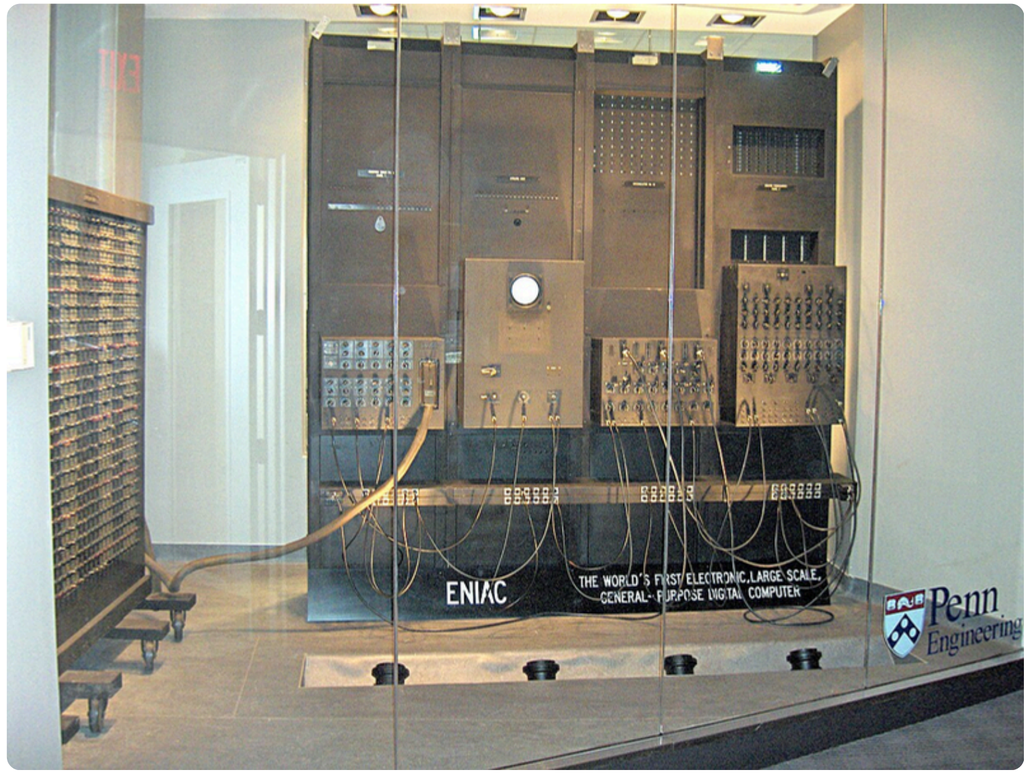


Difference Engine



Analytical Engine

First Electronic Computer



The first computer, the **ENIAC (Electronic Numerical Integrator and Computer)**, was developed in the 1940s and is often considered the first general-purpose electronic digital computer.

It **filled an entire room**, and although primitive by today's standards, the ENIAC was a groundbreaking achievement that laid the foundation for modern computing.

Further Reading

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- **tutorialspoint - Basics of Computers**
  [https://www.tutorialspoint.com/basics_of_computers/index.htm]
- **Wikipedia - Computer**
  [https://en.wikipedia.org/wiki/Computer]
- **CDAC - History of Computers**
  [https://static.vikaspedia.in/media/files_en/education/Digital%20Litercy/TheHistoryofComputers1.pdf]

Feedback

Now it's time to give your valuable feedback

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Thank You!